

Correction TP01**Matière : ATELIER DEVELOPPEMENT MOBILE AVANCE****Classe : SEM31**

Contact

package com.gestion.contact

```
data class Contact(  
    var id: Int,  
    var nom: String,  
    var prenom: String,  
    var numero: String  
) {  
    override fun toString(): String {  
        return "$nom $prenom($numero)"  
    }  
}
```

SQLiteContact

package com.gestion.contact

```
import android.content.Context  
import android.database.sqlite.SQLiteDatabase  
import android.database.sqlite.SQLiteOpenHelper  
  
class SQLiteContact(  
    context: Context?,  
    name: String?,  
    factory: SQLiteDatabase.CursorFactory?,  
    version: Int  
) : SQLiteOpenHelper(context, name, factory, version) {  
  
    override fun onCreate(db: SQLiteDatabase) {  
        val sql = "CREATE TABLE contact (id INTEGER PRIMARY KEY AUTOINCREMENT, " +  
            " nom TEXT NOT NULL, " +  
            " prenom TEXT NOT NULL, numero TEXT NOT NULL);"   
        db.execSQL(sql)  
    }  
  
    override fun onUpgrade(db: SQLiteDatabase, oldVersion: Int, newVersion: Int) {  
        // Laisser vide ou ajouter la logique de mise à jour de la BDD  
    }  
}
```

MainActivity

package com.gestion.contact

```
import android.content.Intent
import android.database.Cursor
import android.database.sqlite.SQLiteDatabase
import android.os.Bundle
import android.widget.AdapterView
import androidx.activity.result.contract.ActivityResultContracts
import androidx.appcompat.app.AppCompatActivity
import androidx.core.widget.doAfterTextChanged
import com.gestion.contact.databinding.ActivityMainBinding

class MainActivity : AppCompatActivity() {
    private lateinit var amb: ActivityMainBinding
    private lateinit var adpContact: ArrayAdapter<Contact>
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        initGraphique()
    }
    private fun initGraphique() {
        amb = ActivityMainBinding.inflate(layoutInflater)
        setContentView(amb.root)
        adpContact = ArrayAdapter(this, android.R.layout.simple_list_item_1)
        amb.lstContact.adapter = adpContact
        remplir()
        ajouterEcouteurs()
    }
    private fun ajouterEcouteurs() {
        amb.btnAjout.setOnClickListener { ajouter() }
        amb.btnSuppression.setOnClickListener { supprimer() }
        amb.lstContact.setOnItemLongClickListener { _, _, position, _ ->
            modifier(position)
            false
        }
        amb.edFiltre.doAfterTextChanged {
            remplir()
        }
    }
    private fun remplir() {
        val b = SQLiteContact(this, "Contact.db", null, 1)
        val db : SQLiteDatabase = b.writableDatabase
        val sql = "Select * From contact order by nom,prenom asc"
        val c: Cursor = db.rawQuery(sql, null)
        adpContact.clear()
        while (c.moveToNext()) {
            val id = c.getInt(0)
            val nom = c.getString(1)
            val prenom = c.getString(2)
            val numero = c.getString(3)
            val contact = Contact(id, nom, prenom, numero)
            adpContact.add(contact)
        }
        c.close() // Pensez à fermer le curseur manuellement
        amb.tvNbContacts.text = adpContact.count.toString()
    }
    private val launcher = registerForActivityResult(
        ActivityResultContracts.StartActivityForResult()
    ) { result ->
        if (result.resultCode == RESULT_OK) {
            remplir()
        }
    }
}
```

```

    }
}
private fun ajouter() {
    val intent = Intent(this, AjoutActivity::class.java)
    launcher.launch(intent)
}

private fun supprimer() {
    val intent = Intent(this, SuppressionActivity::class.java)
    launcher.launch(intent)
}

private fun modifier(position: Int) {
    val c = adpContact.getItem(position)

    if (c != null) {
        val i = Intent(this, ModificationActivity::class.java).apply {
            putExtra("id", c.id)
            putExtra("nom", c.nom)
            putExtra("prenom", c.prenom)
            putExtra("numero", c.numero)
        }
        launcher.launch(i)
    }
}

private fun remplirFiltre() {
    val filtre = amb.edFiltre.text.toString()
    val b = SQLiteContact(this, "Contact.db", null, 1)
    val db = b.writableDatabase

    val sql = "Select * From contact Where nom like ? or prenom like ? or numero like ? " +
        "order by nom,prenom asc"
    val filtreParam = "%$filtre%"

    val c: Cursor = db.rawQuery(sql, arrayOf(filtreParam, filtreParam, filtreParam))

    adpContact.clear()
    while (c.moveToNext()) {
        val id = c.getInt(0)
        val nom = c.getString(1)
        val prenom = c.getString(2)
        val numero = c.getString(3)

        val contact = Contact(id, nom, prenom, numero)
        adpContact.add(contact)
    }
    c.close() // Pensez à fermer le curseur manuellement

    amb.tvNbContacts.text = adpContact.count.toString()
}
}

```

AjoutActivity

package com.gestion.contact

```
import android.content.ContentValues
import android.os.Bundle
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
import com.gestion.contact.databinding.ActivityAjoutBinding

class AjoutActivity : AppCompatActivity() {
    private lateinit var amb: ActivityAjoutBinding
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        initGraphique()
    }
    private fun initGraphique() {
        amb = ActivityAjoutBinding.inflate(layoutInflater)
        setContentView(amb.root)
        ajouterEcouteurs()
    }
    private fun ajouterEcouteurs() {
        amb.btnAjouter.setOnClickListener { ajouter() }
        amb.btnAnnulerA.setOnClickListener { annuler() }
    }
    private fun annuler() {
        setResult(RESULT_CANCELED)
        finish()
    }
    private fun ajouter() {
        if (amb.edNom.text.isNotBlank() &&
            amb.edPrenom.text.isNotBlank() &&
            amb.edNumero.text.isNotBlank()
        ) {
            val b = SQLiteContact(this, "Contact.db", null, 1)
            val db = b.writableDatabase
            val v = ContentValues().apply {
                put("nom", amb.edNom.text.toString())
                put("prenom", amb.edPrenom.text.toString())
                put("numero", amb.edNumero.text.toString())
            }
            db.insert("contact", null, v)
            db.close()
            setResult(RESULT_OK)
            finish()
        } else {
            Toast.makeText(
                this, "Remplir tous les champs SVP!",
                Toast.LENGTH_LONG
            ).show()
        }
    }
}
```

SuppressionActivity

```
package com.gestion.contact
import android.database.Cursor
import android.os.Bundle
import android.widget.AdapterView
import androidx.appcompat.app.AppCompatActivity
import com.gestion.contact.databinding.ActivitySuppressionBinding
class SuppressionActivity : AppCompatActivity() {
    private lateinit var amb: ActivitySuppressionBinding
    private lateinit var adpContact: ArrayAdapter<Contact>
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        initGraphique()
    }
    private fun initGraphique() {
        amb = ActivitySuppressionBinding.inflate(layoutInflater)
        setContentView(amb.root)
        adpContact = ArrayAdapter(this, android.R.layout.simple_spinner_item)
        amb.spContact.adapter = adpContact
        remplir()
        ajouterEcouteurs()
    }
    private fun ajouterEcouteurs() {
        amb.btnSupprimer.setOnClickListener{supprimer()}
        amb.btnAnnulerS.setOnClickListener{annuler()}
    }
    private fun remplir() {
        val b = SQLiteContact(this, "Contact.db", null, 1)
        val db = b.writableDatabase
        val sql = "Select * From contact order by nom,prenom asc"
        val c: Cursor = db.rawQuery(sql, null)
        adpContact.clear()
        while (c.moveToNext()) {
            val id = c.getInt(0)
            val nom = c.getString(1)
            val prenom = c.getString(2)
            val numero = c.getString(3)
            val contact = Contact(id, nom, prenom, numero)
            adpContact.add(contact)
        }
        c.close() // Pensez à fermer le curseur manuellement
    }
    private fun annuler() {
        setResult(RESULT_CANCELED)
        finish()
    }
    private fun supprimer() {
        val c = amb.spContact.selectedItem as Contact
        val id = c.id
        val b = SQLiteContact(this, "Contact.db", null, 1)
        val db = b.writableDatabase
        db.delete("contact", "id=?", arrayOf(id.toString()))
        db.close()
        setResult(RESULT_OK)
        finish()
    }
}
```

ModificationActivity

package com.gestion.contact

```
import android.content.ContentValues
import android.database.sqlite.SQLiteDatabase
import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity
import com.gestion.contact.databinding.ActivityModificationBinding
```

```
class ModificationActivity : AppCompatActivity() {
    private lateinit var amb: ActivityModificationBinding
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        initGraphique()
    }

    private fun initGraphique() {
        amb = ActivityModificationBinding.inflate(layoutInflater)
        setContentView(amb.root)
        remplirChamps()
        ajouterEcouteurs()
    }

    private fun remplirChamps() {
        amb.edNomM.setText(intent.getStringExtra("nom"))
        amb.edPrenomM.setText(intent.getStringExtra("prenom"))
        amb.edNumeroM.setText(intent.getStringExtra("numero"))
    }

    private fun ajouterEcouteurs() {
        amb.btnModifier.setOnClickListener { modifier() }
        amb.btnAnnulerM.setOnClickListener { annuler() }
    }

    private fun modifier() {
        val id = intent.getIntExtra("id", 0)
        val b = SQLiteContact(this, "Contact.db", null, 1)
        val db: SQLiteDatabase = b.writableDatabase
        val v = ContentValues().apply {
            put("nom", amb.edNomM.text.toString())
            put("prenom", amb.edPrenomM.text.toString())
            put("numero", amb.edNumeroM.text.toString())
        }
        db.update("contact", v, "id=?", arrayOf(id.toString()))
        db.close()
        setResult(RESULT_OK)
        finish()
    }

    private fun annuler() {
        setResult(RESULT_CANCELED)
        finish()
    }
}
```